

YASH BAKSHI

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EDUCATION

Indian Institute of Science (IISc)

M.Tech (Research), Computer Science and Automation

Aug 2024 – Present

CGPA: **7.5/10**

Government Engineering College, Raipur

Bachelor of Technology, Computer Science and Engineering

Aug 2019 – May 2023

CGPA: **8.92/10**

PROJECTS

AI Research Assistant using Retrieval-Augmented Generation

LangChain, LLMs, Vector DBs

- Built a research-paper QA system using TF-IDF + SentenceTransformer embeddings, FAISS vector search, Reciprocal Rank Fusion, and cross-encoder reranking, with local Ollama LLM generation and source-grounded citations.
- Developed an automated retrieval evaluation pipeline for multi-PDF corpora, benchmarking sparse, dense, hybrid, and reranked retrieval using Recall@K, MRR, Hit@K, and nDCG@K.

Sentiment Analysis using Deep Learning techniques

PyTorch, Gensim, Scikit-learn

- Built a sentiment classification pipeline on 50K IMDB reviews using 100-dimensional Word2Vec embeddings, stacked LSTMs, attention pooling, and a Transformer encoder.
- Evaluated average-pooling LSTM (**89.46%**), attention-pooling LSTM (**90.22%**), and Transformer (**88.28%**), with the attention model achieving the best **0.901 F1-score**.

Transformer Language Model from scratch

PyTorch, NumPy

- Implemented a Transformer from scratch using PyTorch primitives, including multi-head attention, positional encoding, masking, residual connections, and layer normalization.
- Trained and evaluated the model on [dataset/task], achieving [result] while validating attention shapes, causal masking, and autoregressive generation.

English-to-Hindi Neural Transliteration System

PyTorch, NumPy, NLTK

- Designed a custom BPE tokenizer and trained LSTM and attention-based GRU encoder-decoder models for transliterating Indian names into Hindi (Devanagari).
- Implemented greedy and beam-search decoding with length penalty, achieving **45.25%** exact-match accuracy, **14.26%** character error rate, and a **0.7049** BLEU score.

Credit Card Fraud Detection using Machine Learning

Scikit-learn, XGBoost, SMOTE

- Performed data preprocessing, feature scaling, and **SMOTE**-based class balancing, trained and benchmarked **6 machine learning models** using **5-fold cross-validation** and GridSearchCV.
- Evaluated models using **AUROC**, **AUPRC**, Precision, Recall, and F1-score, achieved best balanced performance with **Random Forest** (AUROC: **0.9711**, F1: **0.7973**) and highest **AUPRC** of **0.8428** using **XGBoost**.

JOB EXPERIENCE

Amicus Technology

Jul 2023 – Apr 2024

Associate Software Analyst

- Developed and deployed scalable enterprise web applications using JavaScript, SQL, and ASP.NET MVC architectures.
- Optimized relational database schemas and data extraction pipelines, enhancing system data retrieval efficiency.
- Maintained production stability of a Logsheet Management Application by debugging and resolving critical system defects.

ACHIEVEMENTS

- All India Rank **577** in GATE CS (2024) among 123,967 candidates
- Rank **87** in CGPET (Chhattisgarh Engineering Entrance Exam)